

The Indian Market for Diodes and Transistors

The Indian market for diodes is expected to undergo a major shift towards chip-based diodes. This has driven some of the leading players to revamp their existing facilities to cater to this demand for chip-based diodes. In transistors, on the other hand, IGBTs and MOSFETs are expected to grow at a faster rate than other transistors



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The Indian electronic components market is dominated by electromechanical components (relays, switches, connectors, etc) and passive components (wound components, resistors, capacitors, etc). However, in recent times, active components have also witnessed significant growth.

Active components are made of semiconducting material and include integrated circuits, diodes and transistors.

Though diodes and transistors currently form a small portion of the active components market in India (refer Fig. 1), the importance of this market cannot be undermined. In this month's industry focus, before we take a closer look at the diodes and transistors markets in India, let us understand the stages involved in diodes' and

transistors' manufacture and India's capability.

Prime requirements for diodes and transistors manufacturing

Generally, diodes/transistors manufacturing can be divided into two distinct phases: front end or chip manufacturing, and back end, where the chips are packaged. This definition is applicable to overall semiconductor manufacturing. The requirements of front-end and back-end manufacture from infrastructure, technology and skill requirement point of view are essentially unique and very different from each other.

Praveen Atnoorkar, director-operations, Vishay Semiconductor India, explains, "Most critical electrical parameters of diodes are related to chip design and are 'fixed' during the manufacturing stages. Several technological inputs are required at the time of processing as well as chip designing. In addition to it, manufacturing processes are required to be carried out in a controlled environment (wafer fabs). Optimisation of manufacturing pro-

cesses is a continuous process. Support of scale of manufacturing is extremely critical from cost management point of view."

Talking about the scenario in India, Atnoorkar adds, "There are very few wafer fabs located in India. Compared to fabs in other countries, these fabs are less efficient in cost management because the scale of manufacturing in India is not favourable. Wafer processing predominantly involves chemical and thermo mechanical processes and requires automation efforts. Indian wafer processing today very much depends on imports of basic silicon, chemicals and also equipment. Therefore it continues to be less efficient than the competition across globe. Today, Taiwan, China and the US have created overcapacity for front-end manufacturing and any new players in India or anywhere will face issues of achieving scale of economy in the near term.

"On the other hand, in back-end manufacturing, or packaging of chips, the processes are more mechanical and designers need to remain focused on optimisation of packaging materials and manufacturing processes with scale of economy. The technological inputs are less dominant compared to front end. The design changes are not often due to standardisation of package outlines. Again, due to absence of many domestic players in India, we are still dependent on the imports of materials. Logistics costs associated with them are critical part of supply chain management of back-end manufacturing in India."

To counter the situation, Atnoorkar suggests the Indian government a two-step process: "First, increase the back-end or assembly houses in India. This can be achieved by providing incentives to component users that are sourcing components built in India. Focus on logistics infrastructure and simplification of taxing structure would help. This is generally unseen part of back-end manufacturing growth. Then after some years, when sufficient back-end production

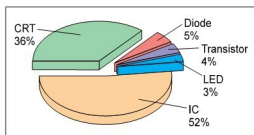


Fig. 1: Active components' market share (Source: ELCINA's ELCOMOS report)